brain. Researchers from the University of British Columbia and the University of Victoria observed 24 healthy adults before and after exposure to pollutants using functional magnetic resonance imaging. Their findings were published in the journal *Environmental Health* in November 2023, and showed that even two hours of exposure to diesel pollution decreases the functional connectivity of the brain. This manifests mostly in poor memory recall. And we are exposed to diesel pollution regularly in traffic conditions.

In October, the Barrow Neurological Institute in Arizona found air pollution to increase the risk of Parkinson's disease by 56 per cent.



"VIRUS OR PM, YOUR BODY REACTS TO FOREIGN BODIES IN THE SAME WAY. THERE IS INFLAMMATION"

DR ARJUN KHANNA
HoD, Pulmonary Medicine, Amrita Hospitals, Faridabad

Chidren have it worse. A June 2023 Keck School of Medicine of University of Southern California study that looked at brain scan data of 9,000-plus youth found that certain pollutants considered safe by the US Environmental Protection Agency are linked to changes in brain function over time. This follows on the heels of a 2022 study published in the journal *Science Advances*, which showed that a rise in exposure to air toxins in infants causes them to fall behind in academics later; for some, the effect is equivalent to not attending an entire month of elementary school. "I cannot stress this enough," says Dr Khanna. "If you think bursting crackers is a source of joy for your children, it is a direct cause of harm to them."

POLLUTION AND CANCER

There is now a growing body of evidence to link air pollution with cancer. A study led by the Harvard T.H. Chan School of Public Health in the US and published in *Environmental Epidemiology* in August 2023 found that chronic exposure to fine particulate air pollutants and nitrogen dioxide raises the risk of four types of non-lung cancers in cancer-free older adults. A cohort study of 2.2 million to 6.5 million (depending on the type of cancer studied) people above the age of 65 (all of whom had been free of cancer for at least 10 years before the period of study) found that exposure over a 10-year period increased the risk of developing colorectal and prostate cancers. Even low levels of exposure to air pollutants were found to make people particularly susceptible to developing these cancers, in addition to breast and endometrial cancers.

Another breakthrough study in September 2022 by the European Society of Medical Oncology found air pollution to cause cancer of the lung. Examining data from over 400,000 European and Asian people, it found that inflammation caused by pollutants awakens in the lungs normally inactive cells that carry cancer-causing mutations. Ongoing inflammation can trigger uncontrollable growth of these cells, forming cancerous tumours. "I get so many patients with lung cancer who say, 'I never smoked. How can I have lung cancer?' Well, in Delhi, you are smoking every single time you take a breath," says Dr Khanna.

A third study, published in 2016 by the American Association of Cancer Research, shows the risk of cancer mortality when you breathe toxic air. It studied 66,280 residents of Hong Kong, and showed that for every 10 µg/m3 of increased exposure to PM2.5, the risk of dying from any cancer rose by 22 per cent. For cancers of the upper digestive tract, the mortality risk was 42 per cent higher; for cancers of the accessory digestive organs—be it the liver, bile ducts, gall bladder or the pancreas—the mortality risk 35 per cent higher. For breast cancer, the mortality risk was 80 per cent higher, and 36 per cent higher for lung cancer. Right now, the current exposure to PM2.5 in Delhi is four times the limit used by the researchers (42 µg/m3 as on November 8, according to IQair.com). But how does air pollution cause death by cancer? Among the explanations given in the study is that PM2.5 can spark defects in DNA repair function, alterations in the body's immune response or inflammation that triggers angiogenesis or the growth of

